Application for Undergraduate final year research Project
Faculty of Computing

APPLICATION FOR UNDERGRADUATE FINAL YEAR RESEARCH PROJECT

<table>
<tr><td colspan="2">1. Project Title
The title of the project should be brief, reflect concisely and accurately the proposed. Students are advised to avoid titles which convey a distant or potential application of the proposed work, or a greater aspiration or goal than is to be expected from the proposed work.</td></tr>
<tr><td colspan="2">IoT-Enabled Smart Waste Collection System with Community Sharing and Real-Time Route Optimization</td></tr>
<tr><td>2. Research Area</td><td>Specific Area</td></tr>
<tr><td>Smart Waste Collection and IoT based smart bin</td><td>Real Time Tracking, Route Optimization, Community Sharing</td></tr>
<tr><td colspan="2">3. Supervisor(s)</td></tr>
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Area of expertise related to the proposed project:</td><td>Mobile :

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5. Summary

(a) Explain briefly the research problem, research approach and expected outputs.
Not exceeding one A4 page, font type Times New Roman, size 11, single space.

Efficient waste collection is a significant challenge in urban areas, leading to environmental pollution, health risks, and logistical inefficiencies due to the lack of real-time monitoring, optimized collection routes, and effective user interaction. This research proposes a Smart Waste Collection System that integrates IoT, Algorithm-based methods and web/mobile applications to address these issues. IoT-enabled bins will monitor waste levels and send real-time updates, while algorithm-based methods will optimize collection routes, reducing fuel consumption and operational costs. A mobile and web application will facilitate real-time tracking of bins and trucks, notify users, and enable community participation through features like reusable item sharing. The system will also include waste segregation guidance, citizen reporting, and multi-language support to enhance usability. Expected outcomes include a scalable solution for efficient waste collection, improved operational transparency, reduced resource wastage, and comprehensive analytics for data-driven decision-making, ultimately promoting a smarter and more sustainable approach to urban waste management.

(b) Give 3 – 5 keywords for the proposed project:

Waste collection, Route Optimization, Real-Time Tracking, IoT Smart Bin, Community Sharing

6. Research Problem

6.1 Research problem/s

- Inefficiency in waste collection processes leads to delayed pickups and uncollected waste.
- Absence of real-time tracking for waste bins and trucks, causing poor scheduling and inefficiencies.
- Inefficient collection routes result in increased fuel consumption and environmental pollution.
- Limited public engagement in waste segregation, reporting, and recycling efforts.
- Lack of integration of advanced technologies like IoT for optimizing operations.
- Challenges in recycling and reusing waste due to the absence of community-driven solutions.
- Limited efforts to promote awareness and sustainable waste management practices.

6.2 Analysis of the problem/s & rationale for the research question

- Inefficiency in Waste Collection Processes:
  Delayed pickups and uncollected waste result in operational inefficiencies and public dissatisfaction. The proposed system will optimize scheduling and improve resource allocation using algorithm-based methods.

- Absence of Real-Time Monitoring:
  Without IoT-enabled sensors, waste bins overflow, and truck locations cannot be tracked, causing irregularities in collection. IoT-based monitoring will provide real-time updates, ensuring timely and efficient operations.

- Environmental Impact of Inefficient Routes:
  Unoptimized routes increase fuel usage and emissions, negatively impacting the environment. Route optimization algorithms powered by algorithm-based methods will reduce travel distances, fuel consumption, and CO_2 emissions.

- Limited Public Engagement in Waste Collection:
  Citizens are not actively participating in waste reporting or segregation due to lack of tools. The introduction of a mobile app with features like reporting issues and sharing reusable items will enhance community involvement.

- Underutilization of Advanced Technologies:
  Traditional systems are not leveraging algorithm-based methods and IoT for smarter operations. Integrating these technologies will modernize the waste collection process, making it efficient, scalable, and adaptable.

- Challenges in Recycling and Reusing Waste:
  Absence of community-driven solutions results in reusable items being discarded. A feature enabling citizens to upload reusable items for sharing can promote recycling and reduce waste.

- Difficulty in Promoting Awareness and Sustainability:
  Lack of awareness about waste segregation and sustainability hinders proper waste collection. The system will provide educational content and awareness campaigns to encourage environmentally friendly practices.

7. Comprehensive literature review AND the complete list of references in the relevant area.

(Attach additional sheets if necessary)

The literature review explores the advancements in smart waste collection systems, emphasizing the integration of modern technologies such as IoT, AI, and mobile applications. It examines existing research on real-time monitoring of waste bins, route optimization for collection trucks, and community-driven solutions to promote sustainability. By analyzing previous studies, this review identifies gaps in current systems and highlights innovative approaches to enhance waste collection efficiency, reduce environmental impact, and improve user engagement. This foundation sets the stage for designing a comprehensive, technology-driven solution tailored to address the challenges of waste collection.

The integration of Internet of Things (IoT) technology into waste management systems has emerged as a transformative approach to addressing the challenges of urban waste collection. Recent studies, including the work by Zeb et al. [1], highlight the potential of smart waste management systems to enhance operational efficiency and environmental sustainability. These systems utilize sensors embedded in waste bins to monitor fill levels in real-time, enabling timely collection and reducing the risk of overflow, which poses hygiene and aesthetic concerns in urban areas. The proposed architecture in the study emphasizes the importance of effective data communication between sensor nodes and a central management system, utilizing wireless technologies to facilitate seamless information flow. Furthermore, the optimization of collection routes based on real-time data not only minimizes operational costs but also contributes to reduced carbon emissions, aligning with global sustainability goals. The research underscores the significance of minimizing end-to-end delays in data transmission, which is critical for ensuring prompt responses to waste collection needs. As cities continue to grow, the implementation of smart waste collection systems represents a vital step towards creating cleaner, more efficient urban environments, ultimately enhancing the quality of life for residents.

Research by Haque et al. [2], highlights the development of an IoT-based waste collection system featuring smart bins that monitor fill levels, odor, humidity, and temperature in real-time. This system enables waste management authorities to identify which bins require emptying, thereby optimizing collection routes and reducing travel distances by an average of 30.76% compared to traditional methods. Additionally, the use of cloud technology allows for remote data access, facilitating informed decision-making regarding collection schedules and resource allocation. The study also emphasizes the importance of energy efficiency in battery-powered IoT devices, addressing a critical challenge in the implementation of smart waste management solutions. Overall, these advancements illustrate the potential of IoT to create more efficient urban waste management systems.

Recent advancements in IoT technology have significantly impacted waste management systems, offering innovative solutions to traditional challenges. Che Soh and Husa et al. [3], proposed the Smart Waste Collection Monitoring and Alert System (SWCMAS) exemplifies this integration by utilizing IoT to enhance waste collection efficiency. As highlighted by recent studies, IoT platforms like Ubidots provide a robust framework for real-time monitoring and data visualization, enabling timely interventions in waste management processes. The system employs ultrasonic sensors interfaced with Arduino UNO to measure waste levels, transmitting this data to the cloud for analysis and alert generation. This method not only optimizes collection routes but also minimizes environmental pollution by preventing bin overflow. The literature underscores the importance of such systems in urban settings, where efficient waste management is crucial for public health and environmental sustainability. By reducing unnecessary collection trips, SWCMAS contributes to resource conservation and operational cost savings, aligning with broader sustainability goals.

Folianto and Sheng Low et al. [4], introduced the Smart bin system, a pioneering approach to waste management that leverages wireless sensor networks to monitor litter bin fullness. The system is designed to optimize waste collection by providing real-time data on bin utilization and daily seasonality. This data is transmitted through a wireless mesh network, employing a duty cycle technique to minimize power consumption and extend operational time. The Smart bin architecture includes outdoor sensor nodes, analytics, and a workstation interface, which collectively enhance the efficiency of waste management operations. Over a six-month test period, the system achieved a 99.25% data delivery ratio, demonstrating its reliability and effectiveness in urban environments. This innovative solution empowers cleaning operators to address cleanliness issues promptly, thereby improving overall productivity and hygiene.

Moreover, in the work of Lozano et al. [5] study a smart waste collection system designed for rural environments, utilizing low-consumption LoRaWAN nodes and route optimization algorithms to improve efficiency. The system employs sensors to measure weight, filling volume, and temperature of waste containers, allowing for real-time monitoring and alerts in case of issues. A dynamic route optimization module is integrated, enabling collection vehicles to minimize distance and operational costs while saving energy and time. A mobile application guides drivers along optimized routes, calculated using data from deployed sensors. The case study conducted in Salamanca, Spain, demonstrated the system’s feasibility, with notable cost reductions and improved resource utilization compared to static route approaches.

The research of Goutam Mukherjee et al. [6], highlights cutting-edge technologies and methods for enhancing waste collection and management to address the pressing global issue of waste pollution. Advanced solutions such as plasma gasification, microbial fuel cells, and bio-refineries are emphasized for their ability to convert waste into energy while minimizing environmental harm. Innovative systems like IoT-enabled smart bins and the Mr. Trash Wheel demonstrate practical applications for real-time waste monitoring and cleanup, offering efficient and sustainable approaches. Additionally, the integration of Geographic Information Systems (GIS) and sophisticated sorting techniques underscores the potential for streamlined, cost-effective waste management. By leveraging successful practices from around the world, this research advocates economic and eco-friendly strategies to tackle the growing challenges of waste disposal and environmental conservation.

The research conducted by Rutqvist et al. [7] introduces an automated machine learning approach for enhancing smart waste management systems, particularly focusing on the detection of container emptying using sensor data. The study explores methods to improve existing manually engineered models by leveraging machine learning techniques, resulting in significant accuracy gains from 86.8% to 99.1%. Key features such as filling levels, vibration strength scores, and time-based data are analyzed to optimize emptying predictions, reducing unnecessary waste transportation and ensuring timely recycling. The findings underscore the critical role of data-driven methodologies and advanced sensors in developing efficient and eco-friendly waste management solutions.

Fatimah et al. [8] propose a cyber-physical system (CPS)-enabled smart waste collection system designed for semi-urban cities in Indonesia. The study integrates physical and cyber spaces to optimize waste management by leveraging sensors, actuators, IoT, and ICT for real-time monitoring and automation. The CPS model improves the efficiency of waste collection, sorting, and reporting while promoting sustainability. A case study in Magelang highlighted the system's ability to reduce resource use, enhance traceability, and provide transparency throughout the waste collection lifecycle. Challenges such as infrastructure costs, skill gaps, and limited ICT access were also discussed, emphasizing the need for community engagement and government support for successful implementation.

Afiata Donny et al. [9] highlight the critical challenges posed by e-waste, including hazardous material content, recycling inefficiencies, and the need for effective disposal systems. This research emphasizes that addressing these issues requires innovative solutions, such as user-centered mobile applications, bridge the gap between households, collectors, and recycling facilities. By leveraging Design Thinking methodologies, which focus on empathy, ideation, prototyping, and testing, user interfaces can be tailored to meet specific user needs effectively. This approach ensures usability and functionality, enabling more efficient e-waste collection and management systems.

Mukherjee et al.[10] explored sensor-driven smart bins that provide waste level data to optimize collection routes. These bins are integrated with Geographic Information Systems (GIS), allowing municipalities to leverage spatial data in route planning. GIS applications aid in efficiently mapping out collection paths, ensuring that resources are allocated effectively. The sensor data enables a systematic approach to route planning, cutting down on fuel consumption and reducing overall operational costs.

Real-time data collection and predictive analytics are valuable in improving waste collection efficiency. C. S. de Morais and T. R. Ramos [11] introduced a data-driven dynamic routing framework that incorporates machine learning models, such as Hidden Markov Models (HMMs), to predict waste generation and optimize collection routes. Their model, the Data-Driven Dynamic Robust Internet Distributed Routing Problem (D-DRIRP), leverages historical data to enhance route planning and waste bin filling forecasts. This system uses a rolling horizon strategy to adjust collection schedules daily, reducing overflow situations and enabling more efficient resource allocation.

Abd Wahab et al.[12] addressed Malaysia's low recycling rate of 5%, significantly below the 22% target for 2020. The authors highlighted the need for effective strategies to promote the 3R principles (reduce, reuse, recycle) and discussed the use of different technologies to enhance recycling efficiency through real-time waste tracking. The proposed smart recycles bin features a reward system that incentivizes users to accumulate points for recycling, encouraging community participation and raising awareness. The authors advocated further research and collaboration among stakeholders to improve recycling rates and waste management practices, emphasizing the potential of smart recycling systems to tackle these challenges.

Further, Mager and Blass [13], explored the potential of drone technology and advanced data analysis tools in transforming illegal waste dumps into beneficial resources. Their study highlighted the feasibility of using drones for mapping and analyzing illegal waste sites, focusing on the economic and environmental benefits of material recovery. The authors presented pilot data that suggests significant cost reductions in cleanup efforts, estimating potential savings of up to 58 million USD and profits of 82.8 million USD from recovered materials. Additionally, the research connected its findings to the Sustainable Development Goals (SDGs), emphasizing the role of effective waste management in promoting a circular economy. Despite the promising results, the study acknowledged challenges such as data accuracy and the need for further automation in waste identification processes, paving the way for future advancements in sustainable waste management practices.

The findings from these studies, the integration of IoT, AI, GIS, and other innovative technologies presents a transformative opportunity to enhance waste collection systems. Addressing challenges such as cost, data security, and operational complexity is essential for realizing the full potential of these solutions. These findings lay the foundation for designing comprehensive, technology-driven systems to optimize waste collection processes and promote environmental sustainability.

References

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[9] "Development of User Interface Design for Electronic Waste Collection with a Design Thinking Approach," in *Proceedings of the International Conference on Industrial Engineering and Operations Management*, Orlando, USA: IEOM Society International, Jun. 2022, pp. 1636–1646. doi: 10.46254/NA07.20220383.

[10] A. Goutam Mukherjee *et al.*, "A review on modern and smart technologies for efficient waste disposal and management," *J. Environ. Manage.*, vol. 297, p. 113347, Nov. 2021, doi: 10.1016/j.jenvman.2021.113347.

[11] C. S. de Morais, T. R. P. Ramos, M. Lopes, and A. P. Barbosa-Póvoa, "A data-driven optimization approach to plan smart waste collection operations," *Int. Trans. Oper. Res.*, vol. 31, no. 4, pp. 2178–2208, 2024, doi: 10.1111/itor.13235.

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[13] A. Mager and V. Blass, “From Illegal Waste Dumps to Beneficial Resources Using Drone Technology and Advanced Data Analysis Tools: A Feasibility Study,” *Remote Sens.*, vol. 14, no. 16, Art. no. 16, Jan. 2022, doi: 10.3390/rs14163923.

8. Originality & innovativeness of the proposed work

The proposed development of a Smart Waste Collection System exhibits significant originality and innovation in several key aspects. Firstly, tailoring the system specifically for Sri Lanka acknowledges the unique waste management challenges, environmental concerns, and cultural practices of the country. This localization ensures that the system addresses the specific needs of Sri Lankan communities, setting it apart from generic global waste collection solutions.

Secondly, the use of IoT sensors, AI-driven algorithms, Report generation and prediction, and real-time tracking technologies provides a cutting-edge approach to waste collection. These technologies enable precise and efficient solutions, such as real-time monitoring of bin fill levels, optimized collection routes, and predictive analytics for waste trends. This novel application of advanced technologies has the potential to revolutionize waste collection processes, particularly in areas with resource constraints or inefficient systems.

Furthermore, the emphasis on community engagement and sustainability demonstrates the system’s holistic approach to waste collection. A key innovation within this approach is the community-sharing feature, which empowers users to share reusable items in their neighborhoods. By facilitating this exchange, the system reduces waste generation at its source, promotes the reuse of resources, and encourages a culture of sustainability. This feature not only alleviates pressure on landfills but also strengthens community bonds by fostering collaboration and resource-sharing.

Additionally, features like waste segregation guidance and citizen reporting further enhance the system’s ability to reduce waste and maintain cleanliness. These tools allow users to actively engage in sustainable practices and report issues, ensuring a more efficient and responsive waste management process.

Overall, the proposed work's uniqueness and innovativeness stem from its combination of localized focus, advanced technological integration, and community-centered features, particularly the community-sharing initiative. This makes it a pioneering effort in improving Sri Lanka’s waste collection infrastructure and promoting environmental sustainability.

9. Overall aim and specific objectives of the proposed work

9.1 Overall aim

The overall aim of the proposed Smart Waste Collection System is to develop an innovative, efficient, and sustainable waste collection solution tailored to the specific needs of Sri Lanka. By integrating advanced technologies such as IoT, and real-time tracking, the system seeks to optimize waste collection processes, reduce environmental impact, and foster a culture of sustainability within communities. The system aims to improve the efficiency of waste collection, minimize waste generation, and promote active community participation in maintaining cleanliness and sustainability.

9.2 Specific Objective/s

- Develop a real-time waste bin monitoring system using IoT sensors to optimize collection schedules and prevent overflow.
- Integrate AI algorithms for route optimization, reducing fuel consumption and collection time.
- Create a community-sharing feature for users to share reusable items, reducing waste and promoting sustainability.
- Implement waste segregation guidance to educate users on proper disposal practices.
- Enable citizen reporting to address issues like illegal dumping and overflowing bins.
- Provide multi-language support to ensure inclusivity for all users.
- Enable real-time tracking of garbage trucks for improved transparency and convenience.
- Generate data analytics and reports for authorities to improve waste management efficiency.

10. Methodology

10.1 Describe the Methodology
(Attach additional sheets if necessary)

1. IoT Waste Bin Monitoring: Install IoT sensors in bins to track fill levels and send real-time updates to the system.
2. Route Optimization with Algorithms: Use AI to create optimized routes for trucks based on bin locations and traffic data.
3. Community Sharing Feature: Enable users to share reusable items through a mobile app, reducing waste and promoting sustainability.
4. Waste Segregation Guidance: Provide in-app guidance on proper waste disposal and segregation.
5. Citizen Reporting: Allow users to report issues like illegal dumping or overflowing bins directly through the app.
6. Multi-Language Support: Include multiple language options in the app and web platform for accessibility.
7. Real-Time Truck Tracking: Track garbage trucks live and display their locations for users and administrators.
8. Data Analytics and Reports: Analyze collected data to generate reports for authorities to improve waste collection efficiency.

Tools and Technologies

Hardware Components:

- ESP32: Acts as the central microcontroller in IoT bins and collection trucks, enabling real-time communication and processing data from sensors.
- GPS Module (e.g., NEO-6M): Tracks the real-time location of collection trucks and bins for efficient route planning.
- LoRa Modules (e.g., RFM95W) or BLE: Enable long-range communication between bins and trucks for seamless coordination.

- LEDs: Provide fill levels on bins to indicate statuses like full bins or system errors.

Software and Platforms:

- Flutter: Powers the mobile app that users and drivers use to interact with the system, such as scheduling pickups and viewing notifications.
- React: Builds the admin web panel, providing municipal councils with a user-friendly interface to manage operations.
- Node.js: Handles backend services for processing requests, managing data, and controlling communication between devices.
- Google Maps API: Enables geofencing, real-time route optimization, and map-based visualization for collection trucks and bin locations.
- Firebase: Manages real-time data storage, system alerts, and push notifications for users, drivers, and administrators.

Communication Protocols:

- ESP-NOW: Facilitates wireless communication between IoT devices, such as bins and nearby trucks, for fast data exchange.
- MQTT: Ensures lightweight and reliable messaging for real-time bin status updates and vehicle coordination.
- HTTP/HTTPS: Secures data exchange between the mobile app, admin panel, and backend server for efficient and safe communication.

Version Control

- GitHub: For version control and collaboration.

10.2 Experimental design where applicable
Please complete relevant sections

Smart waste management system where homes and offices generate waste that is segregated into IoT-enabled bins. These bins monitor waste levels and send data to a central database, which organizes optimized routes for collection trucks. A mobile app connects users to the system for scheduling, notifications, and sharing recyclable items within the community. Drivers use the system to collect waste efficiently, while administrators monitor operations through a web-based panel. Reports and analytics help improve processes, promoting recycling, reducing waste, and ensuring seamless coordination.

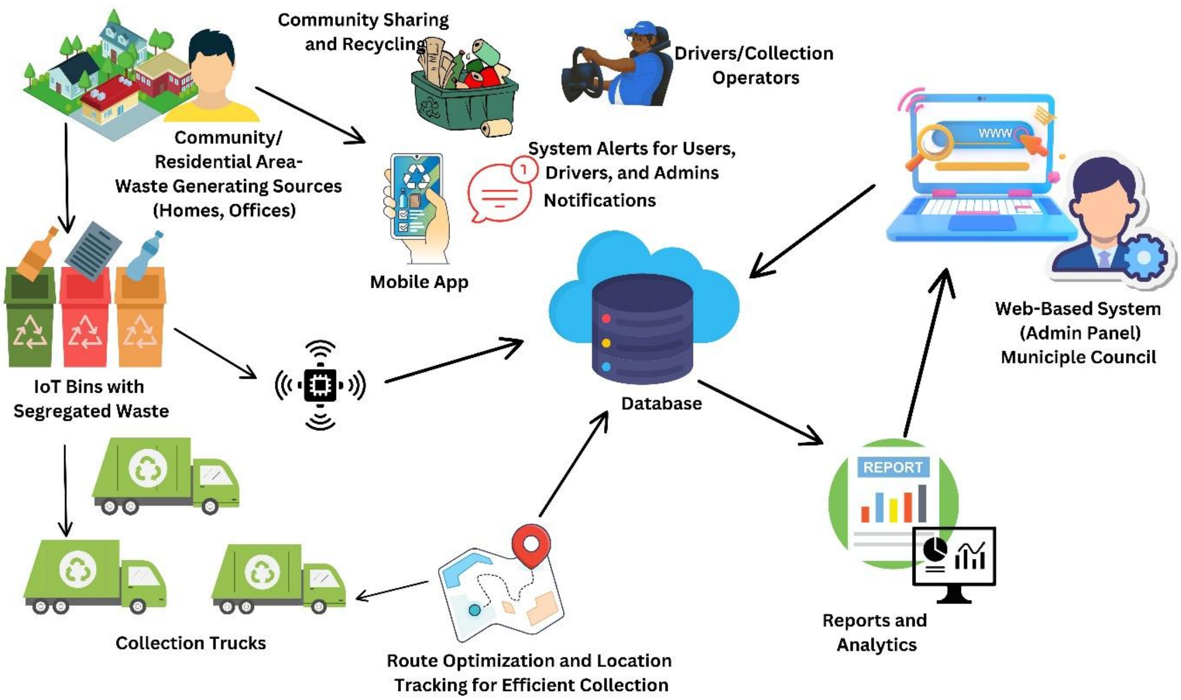
Functional Requirements

1. **User Management**
    - Users (residents, drivers, and administrators) can create, edit, and manage their accounts.
    - Administrators can assign roles, manage permissions, and monitor user activity.
2. **Waste Collection Monitoring**
    - IoT-enabled waste bins detect and transmit real-time waste levels to the central system.
    - Residents can schedule regular waste pickups through the mobile application.
3. **Community Sharing Platform**
    - Enable residents to share reusable items via the mobile app by uploading photos and descriptions.
    - Allow neighbors to claim items within a specified timeframe.
4. **Immediate Pickups**
    - Residents can request immediate waste pickups for urgent situations via the mobile app.
    - Drivers receive real-time notifications of immediate pickup requests and can prioritize these based on their current location and workload.
5. **Route Optimization**
    - The system generates optimized routes for collection trucks based on waste bin locations, fill levels, and immediate pickup requests.
    - Routes are dynamically updated to accommodate emergency pickups or other changing conditions.

6. **Tracking and Mapping**
   - Display bin locations and their statuses on an interactive map for users and administrators.
   - Provide drivers with clear, optimized routes, including immediate pickup locations, on their interface.
7. **Alerts and Notifications**
   - Notify users of scheduled waste pickups, full bins, and immediate pickup statuses.
   - Send real-time alerts to drivers for immediate pickup requests, route changes, or emergencies.
8. **Reports and Analytics**
   - Generate insights on waste collection trends, system performance, and overall efficiency.
   - Provide administrators with detailed weekly or monthly reports, including data on immediate pickups, route efficiency, and user engagement metrics.

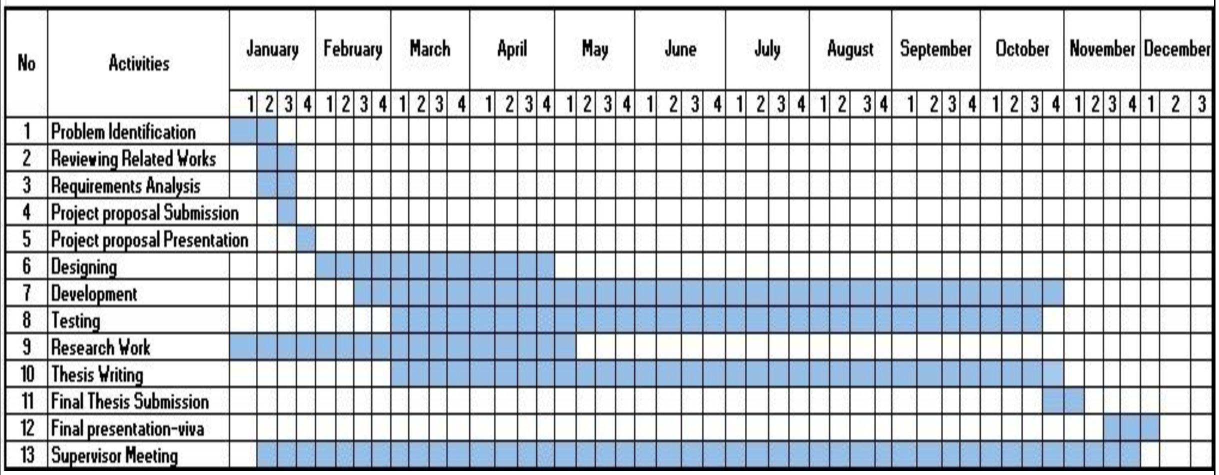
Non-Functional Requirements

1. **Scalability**: Support growth in users, IoT devices, and trucks.
2. **Reliability**: Ensure continuous uptime and system stability.
3. **Performance**: Deliver real-time updates and notifications efficiently.
4. **Security**: Protect data with encryption and controlled access.
5. **User-Friendly**: Provide intuitive and accessible interfaces for all users.



10.3 **Work plan**

Please attach the quarterly Gantt Chart to cover the proposed study, as per the format below.



10.4 **Indicators & milestones of progress**

Please list the milestones and indicators that will be used to measure the progress of the proposed Study

- Completion of research and requirements gathering, finalizing system requirements including IoT, AI, and community features.
- Approval of the prototype design for mobile and web applications.
- Successful integration of IoT sensors, AI algorithms, and real-time tracking technologies.
- Completion of front-end and back-end development for the app’s user interface and infrastructure.
- Successful user testing and quality assurance to ensure system functionality and user experience.
- Deployment and launch of the platform on mobile app stores and web platforms.
- Ongoing platform performance monitoring and collection of user feedback for improvements.
- Achieving desired user acquisition and engagement metrics within the first few months.
- Completion of user training and documentation for effective system use.
- Continuous improvement of the platform based on data analysis and user feedback.

11. Ethical considerations

Relevance to the project	Relevant ☐ Not relevant ☒

12. Plan to protect the human & environmental safety issues related to the project

- Providing appropriate training and instructions on how to use the app and IoT sensors safely and effectively.
- Regularly inspecting and maintaining the IoT sensors, trucks, and other hardware to ensure safe operation.
- Implementing safeguards against data breaches and ensuring data privacy by encrypting user information.
- Following local regulations and safety guidelines for waste management, including environmental and health standards.
- Providing regular maintenance and checks for trucks to ensure environmental safety and fuel efficiency.

- Offering support and resources for users reporting issues or experiencing problems with the system.
- Implementing environmental safety measures, such as promoting waste segregation and reducing landfill waste.

13. Expected Research Outputs

Deliverables at the end of the project in point form

- Fully developed and functioning mobile and web application for waste collection, including features like real-time bin monitoring, route optimization, and community engagement.
- Integrated IoT bin for real-time monitoring of waste bin fill levels, enabling timely collection alerts.
- AI algorithms for optimizing waste collection routes and predicting waste trends based on data.
- User-friendly interface that allows users to track collection schedules, report issues, and participate in community-sharing features.
- Comprehensive database containing waste collection data, including bin fill levels, truck routes, user feedback, and environmental impact metrics.
- Documentation that includes user manuals, technical specifications, and guidance for system implementation and future updates.
- Pilot testing reports evaluating the system’s effectiveness, user engagement, and areas for improvement.
- Regulatory compliance documentation ensuring adherence to local environmental, waste management, and data privacy regulations.
- Deployment of the application to app stores and web platforms for public release and user access.

14. Plan to utilize the research output

- Integrate the system into local waste management practices to optimize waste collection routes, reduce overflow, and improve recycling efforts.
- Conduct outreach and training for local authorities, waste management teams, and communities to familiarize them with the app's functionalities and encourage active participation.
- Implement continuous monitoring and updates to improve the app’s features based on user feedback and operational data.
- Collaborate with environmental and municipal experts to ensure the system’s effectiveness and alignment with local waste management policies.
- Conduct further testing and validation to assess the system’s performance across different regions and waste management contexts.
- Explore opportunities to expand the system to other cities or regions facing similar waste management challenges.
- Support public awareness campaigns promoting sustainable waste management practices, recycling, and waste reduction through the app.

15. Research Outcomes

15.1 Significance of research outcomes and the impact on national/socio-economic development of Sri Lanka.
Please write in point form.

- Improved Waste Management Efficiency: The system will optimize waste collection processes, reducing delays and operational costs, leading to a more efficient waste management system in Sri Lanka.
- Environmental Sustainability: By promoting recycling and waste segregation, the system will contribute to a cleaner environment, reducing the burden on landfills and supporting Sri Lanka's sustainability efforts.
- Economic Growth and Job Creation: The deployment and operation of the system will create job opportunities in technology, waste management, and customer service sectors, boosting the local economy.
- Reduction in Public Health Risks: Timely waste collection and proper waste segregation will help reduce health hazards related to poor waste management, such as the spread of diseases.
- Cost Savings for Local Authorities: The system's route optimization and resource allocation will reduce fuel consumption and operational costs, providing significant savings for municipalities.
- Increased Public Awareness and Engagement: The system encourages active community participation, fostering a culture of environmental responsibility and improving waste management practices at grassroots level.
- Technological Innovation: By incorporating IoT, Sri Lanka will position itself as a leader in innovative waste management solutions, attracting investment and fostering technological advancements.
- Data-Driven Policy Support: The system's data collection and analytics will provide valuable insights to local authorities, helping shape more effective and efficient waste management policies.
- Support for National Development Goals: The project aligns with Sri Lanka's national goals for sustainable development, contributing to the achievement of the United Nations Sustainable Development Goals (SDGs), particularly in environmental protection and urban sustainability.

15.2 Identify relevant stakeholders and potential beneficiaries

- Municipalities: managing waste collection.
- Residents: Citizens benefiting from cleaner environments.
- Government: Regulatory bodies overseeing waste management policies.
- Technology Developers: Engineers and AI specialists developing the system.
- Environmental Organizations: NGOs promoting sustainability.
- Local Businesses: Companies benefiting from a cleaner environment.
- Public Health Officials: Authorities focused on reducing health risks.
- Community Groups: Active users participating in waste segregation.
- Academic Institutions: Universities contributing to research and development.

16. Plan to protect and exploit Intellectual Property (IP)?
(Indicate if applicable)

- Not Applicable

17. Signature of Supervisors

The details given above are correct and personally verified.

Kasunika

(a) ………………………………
(Student)

2025. 01. 24
Date

(b) ………………………………
(Principal Supervisor)

(c) ………………………………
(Co- Supervisor -1)

18. Recommendation and Approval

The application is recommended.

……………………………………………...…… ……………………………..
Department Project Coordinator Date

The application is approved.

……………………………………………...…… …………………………….
Head of the Department Date

Guidelines for Applicants

1. The proposed research project should necessarily be an original investigation.

2. The problem to be tackled should be clearly identified.

3. A comprehensive literature review should be done and all details should be provided together with the list of references.

4. The budget should be calculated correctly and stated in detail.

5. A detailed time-based work plan (Gantt chart) should be included stating the proposed activities and time frames clearly.

6. In the case of selection of a Supervisor from an outside institution, it is mandatory to get an internal supervisor appointed.

7. Softcopy can be obtained from the department in word format. Softcopy of the application should be submitted as PDF format.

8. Incomplete applications **will not** be considered.

TWO hard copies of the completed application form should be submitted to:
Project Coordinator
Department of Computer Science, Faculty of Computing

An electronic version (PDF) should also be e-mailed to hoditm@kdu.ac.lk**, on or before the deadline.**